

General Specification for Civil & Structural Works

Project: Proposed G+4 Residential Building

Date: June 14, 2025

Document Ref: PROJ-SPEC-001

1.0 General

1.1. All work **shall** conform to the latest applicable national standards and codes of practice.

1.2. The Contractor **must** obtain approval for all materials prior to their use in the works.

2.0 Division 03 - Concrete Works

2.1. **Cement:** Cement used **shall** be Ordinary Portland Cement (OPC) grade 53, fresh and free from lumps.

2.2. **Aggregates:** All aggregates **shall** be clean, hard, and durable, and must be washed to remove any impurities. The size and grading of aggregates **shall** be as specified in the approved mix design.

2.3. **Reinforced Cement Concrete (RCC):** The minimum grade of concrete for all RCC work **shall** be M25. The mix **must** be designed by a certified laboratory. Water-cement ratio **is to be** strictly maintained as per the approved design.

2.4. **Formwork:** Formwork **shall** be sufficiently rigid and tight to prevent the loss of grout. It must be cleaned and oiled before every use. Formwork **shall not** be removed until the concrete has achieved sufficient strength.

Table 2.1: Concrete Strength & Testing Requirements

Concrete Grade	Min. Compressive Strength (28 days)	Slump Range (mm)	Testing Frequency
M25	25 N/mm ²	75 - 125	1 set of 3 cubes per 50 m ³
M30	30 N/mm ²	75 - 125	1 set of 3 cubes per 50 m ³

M35	35 N/mm ²	50 - 100	1 set of 3 cubes per 30 m ³ or day
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3.0 Division 05 - Steel Reinforcement

3.1. **Reinforcement Bars:** All reinforcement **shall** be high-yield strength deformed bars (TMT) of grade Fe-500.

3.2. **Bending:** Reinforcement bars **shall** be bent cold to the shapes and dimensions shown on the drawings.

3.3. **Placing:** All reinforcement **must** be placed in the correct position as per the drawings and securely tied to prevent displacement during concreting. A minimum concrete cover **is to be** maintained as per the schedule below.

Table 3.1: Minimum Concrete Cover

Structural Member	Minimum Cover (mm)
Foundations	50
Columns	40
Beams	25
Slabs	20

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